

Teaching notes

This resource contains 12 'standard' trigonometry questions on identifying ratios and sides, and calculating missing values. It also contains one 'exam style' question.

There are two ways this activity could be used. In both activities, students may find they need some paper to show workings.

Option one – treasure hunt

Print all 13 question cards and photocopy onto A3 paper. Cut out each card, shuffle and put up on walls around the classroom, corridor or playground. Arrange students into small groups or pairs, give each group an answer sheet, and send them to a random starting card. Students should read the question and work out the solution, recording their answer on their sheet. They then find the next card in the loop by searching around the room for the card showing their answer. The students should solve the new question and repeat until they end up back at their starting card.

Option two – dominoes

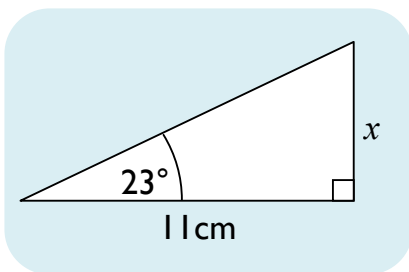
Print the questions onto A4 paper and cut out (you could get the pupils to cut them out as they are not given in the correct order). Students can work individually or in pairs to match up the questions with the correct solutions, creating a loop of dominoes.

Answers

(The cards create a loop so students could start at a different card.)

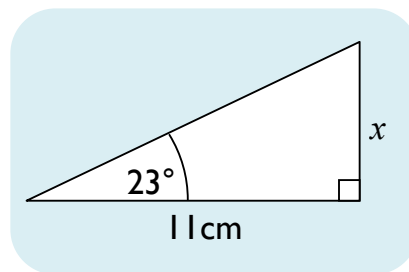
16.1cm → opposite → tangent → 4.7cm → 21.8° → adjacent → cosine → 8.4m → 33.8° → hypotenuse → sine → 15.9cm → 36.9° → 16.1cm

16.1 cm



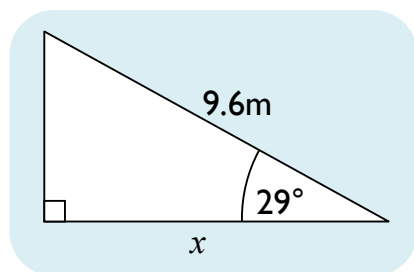
What do we call side x :
opposite, adjacent or
hypotenuse?

opposite



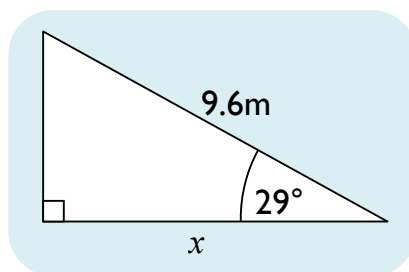
Which ratio would you use to
find x : sine, cosine or tangent?

21.8°



What do we call side x :
opposite, adjacent or
hypotenuse?

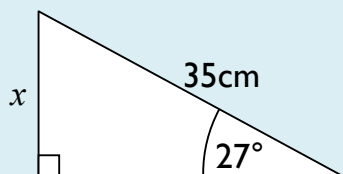
adjacent



Which ratio would you use to
find x : sine, cosine or tangent?

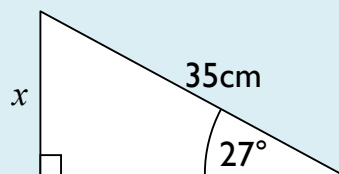


33.8°



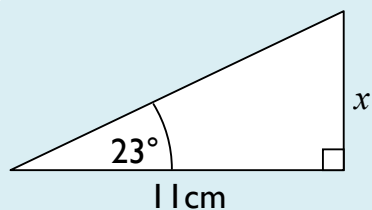
What do we call the side which is 35cm long: opposite, adjacent or hypotenuse?

hypotenuse



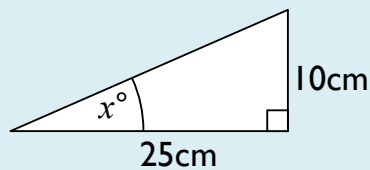
Which ratio would you use to find x : sine, cosine or tangent?

tangent



Calculate the length of x to one decimal place.

4.7cm

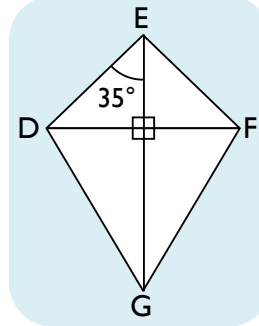
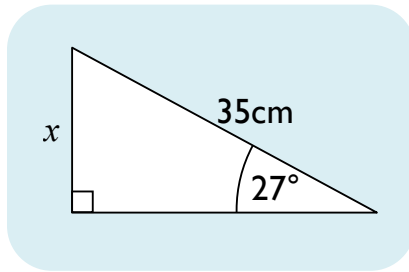


Calculate the size of angle x to one decimal place.



sine

36.9°



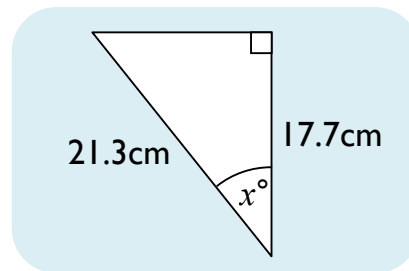
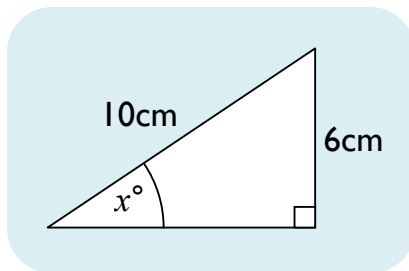
DEFG is a kite. Angle DEG is 35° and line EF is 14cm.

Calculate the length of x to one decimal place.

Calculate the length of DF to one decimal place.

15.9cm

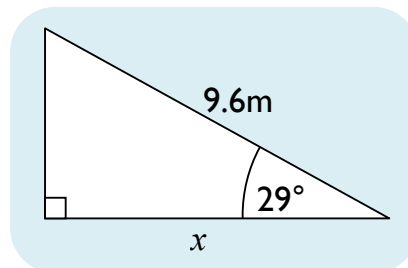
8.4m



Calculate the size of angle x to one decimal place.

Calculate the size of angle x to one decimal place.

cosine



Calculate the length of x to one decimal place.

Answer sheet

Name:

Record your sequence of answers:

